



Longitudinal relationship between the gut microbiota variation and diversity and gut graft-versus-host disease (GVHD) following pediatric allogeneic hematopoietic cell transplantation (HCT) – Case series

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ABSTRACT

Allogeneic Hematopoietic Cell Transplantation (HCT) offers children with life-threatening diseases a chance at survival. Complications from graft-versus-host disease (GVHD, Stages 0–4) represent a significant cause of morbidity and mortality which has been recently associated with gut dysbiosis in the adult HCT population. Here, our objective was to conduct a prospective, longitudinal cohort study in nine pediatric allogeneic HCT participants by collecting longitudinally post-HCT stool specimens up to 1 year. Stool microbiota analyses showed that allogeneic HCT and antibiotic therapy lead to acute shifts in the diversity of the gut microbiota with those experiencing stages 3–4 gut GVHD having significantly greater microbiota variation over time when compared to control participants ($p = 0.007$). Pre-HCT microbiota diversity trended towards an inverse relationship with gut microbiota stability over time, however, this did not reach statistical significance ($p = 0.05$). Future large prospective studies are necessary to elucidate the mechanisms underlying these dynamic changes in the gut microbiota following pediatric allogeneic HCT.

1. Introduction

Allogeneic hematopoietic cell transplantation (HCT) offers children with malignant and non-malignant diseases a chance at survival. Within the first year after HCT, complications from infection and graft-versus-host disease (GVHD) are the leading causes of death; with one often triggering the other (Copelan, 2006). Acute GVHD occurs within the first 100 days following transplantation and is caused by alloreactive donor T cells targeting the recipient's cells leading to cytokine release and parenchymal destruction in organs such as the skin, liver, and gastrointestinal system (gut) (Ghimire et al., 2017). First-line therapy for gut GVHD is systemic glucocorticoids to decrease the inflammatory response, unfortunately, the failure rate is 50 % (McDonald, 2016). Some investigators have looked towards the gut microbiota to determine how bacterial gut microbiota contributes to the underlying immune modulatory mechanisms of GVHD.

Several studies have reported an association between microbial composition and function with morbidity and mortality (reviewed in Ref (Andermann et al., 2018)). Studies in the 1970s demonstrated that decontamination of gut bacteria decreased the risk of gut GVHD (Andermann et al., 2018). However, recently a more nuanced and sophisticated understanding of microbial communities and gut GVHD have emerged. Specifically, depletion of obligate anaerobes (e.g., *Clostridium*) and mono-domination by other bacteria (e.g., *Enterococcus*) are associated with severe gut GVHD (stages 3–4) (Heit et al., 1977; van Bekkum et al., 1967; Veenendaal et al., 1988). Adult patients with reduced intestinal bacterial diversity or mono-domination of organisms such as *Enterococcus* have higher overall mortality rates following allogeneic HCT (Davidovics et al., 2019; van Lier et al., 2020; Vossen et al., 1990; Holler et al.; Stein-Thoeringer et al.). Taur et al. demonstrated that adult patients with lower bacterial diversity profiles in stool 7 days after neutrophil engraftment had a lower survival rate at 3 years

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compared to those with higher intestinal diversity profiles at the same time point (36 % vs. 67 %) (Taur et al., 2014). Recently, Peled et al. showed that lower intestinal diversity was associated with both transplantation-related mortality and gut GVHD-related mortality in adults (Peled et al., 2020). These studies suggest a pivotal role that the gut microbiota plays in determining mortality outcomes after allogeneic HCT.

Pediatric studies on the microbiome and GVHD are limited. Biagi et al. showed that children who experience acute GVHD have significant bacterial gut dysbiosis following their first 3–4 months after allogeneic HCT compared to those without GVHD (Biagi et al., 2015). Additionally, children with GVHD had a lower abundance of the *Bacteroidetes*, a phylum known to produce a propionic acid which has been shown to promote gut integrity and bacterial colonization in naïve hosts (Biagi et al., 2015) and may contribute to the poor or slow mucosal healing seen in GVHD patients (Sofi et al., 2021; Lee et al., 2013). Elgarten et al. found that children undergoing allogeneic HCT demonstrated a pre-HCT stool with a decrease in *Bacteriodes* and *Ruminococcaeae* compared to healthy controls likely due to pre-HCT chemotherapy and/or antibiotic exposures (Elgarten et al., 2022). Additionally, Ingham et al. confirmed the findings by Elgarten et al. by demonstrating that the microbial composition pre-HCT could predict the stage of acute GVHD post-transplantation using a machine learning algorithm, however, longitudinal biospecimens are necessary to confirm these findings. Taken together, pre-HCT stool composition may be a useful predictor of patients who are at the greatest risk of developing severe gut GVHD. Thus, there exists an unmet need to develop a deeper longitudinal understanding of gut microbial profiles in pediatric allogeneic HCT patients which may aid in identify high risk individuals who would benefit from early therapeutic interventions.

The aim of this prospective longitudinal cohort study is to describe the relationship between microbiota prior to HCT and correlating changes in microbial profiles with gut graft-versus-host disease (GVHD) in children undergoing allogeneic HCT. Nine pediatric allogeneic HCT participants had pre-HCT and serial post-HCT stool specimens analyzed using shotgun metagenomic sequencing to identify gut microbiota diversity and longitudinal microbiota variation over time. Defining the relationship between the gut microbiota and gut GVHD remains a key deficiency in our understanding of pediatric allogeneic hematopoietic cell transplantation. Expansion of our knowledge in this area is required to bring novel therapeutic interventions such as fecal microbiota transplantation to this vulnerable patient population.

2. Methods

2.1. Study participants

We conducted a prospective, longitudinal cohort study in pediatric participants who received an allogeneic HCT at the University of California, Los Angeles (UCLA) Mattel Children's Hospital in Los Angeles, California. This study was reviewed and approved by the UCLA Institutional Board Review committee (IRB# 19-000853). Written consent forms were obtained by participants who were 18 years old or by their legal guardians if participants were < 18 years old. Assent forms were obtained from participants who were between 7- to 17- years-old. Enrollment occurred between August 2019 to February 2020 which included institutional clinical research restrictions due the COVID-19 pandemic. Inclusion criteria included children ages of 3 months to 25-years-old and who received an allogeneic HCT. All participants were in HEPA-filtered single rooms and maintained standard hygienic precautions. Antibiotics were given for neutropenic fever, hemodynamic instability or at the discretion of the treating physician based on the participant's clinical course. Antibiotic prophylaxis strategy was not utilized in this patient population. Pneumocystis pneumonia prophylaxis was stopped on Day – 2 and restarted after neutrophil engraftment or Day + 28 whichever came first. Antiviral and antifungal prophylaxis

were administered per institutional standards beginning Day – 1 and Day + 2 respectively.

Following enrollment, participants had stool specimens collected at the following time points: prior to or on the first day of their conditioning regimen (Pre-HCT), weekly until first hospital discharge after HCT and on Days + 60, + 90, + 120, + 150, + 180, + 270 and + 365. Weekly specimens were also collected if participants were admitted to the hospital for any indication. Pre-HCT specimens served as each participant's control. Clinical data including HCT indication, conditioning regimen, GVHD staging along with post-transplantation complications were collected during the same timepoints using the Research Electronic Data Capture (REDCap Consortium; <https://www.project-redcap.org/>) database. Staging for skin, gut and liver is based on the Glucksberg staging criteria with gut GVHD being based on stool output with pediatric-specific staging in mind (stage 0 <10 ml/kg/day, stage 2 10–19.9 ml/kg/day, stage 3 20–30 ml/kg/day and stage 4 >30 ml/kg/day) (Glucksberg et al., 1974; Harris et al., 2016). If an intestinal biopsy was performed, the histological diagnosis was used for staging purposes. GVHD staging is classified as follows: stage 0 being without signs of GVHD, stages 1–2 representing mild GVHD and stages 3–4 representing severe GVHD in order to identify the highest mortality risk patient group (i.e., Stages 3 and 4). Participants were classified as 'controls' if they experienced no or mild forms of gut GVHD (stage 0–2) following allogeneic HCT. Participants were classified as 'gut GVHD' if they experienced the most severe forms of gut GVHD (stages 3–4) at least once following HCT. The initial therapy for GVHD was systemic steroids, however, a combination of other immunosuppressive medications were used including but not limited to basiliximab, budesonide, beclomethasone, extracorporeal photopheresis were individually managed for each participant based on clinical response and clinical presentation.

2.2. Stool specimen collection

Stool specimens were collected at the time points described above. If the participant had a bowel movement within 2 days before or after the intended collection timepoint, the intended time point was used allowing for individual variation in stooling frequency. Stool specimens were collected using sterile stool collection tubes and placed directly on ice and transported to the laboratory. Specimens were aliquoted into several 5 ml cryotubes and maintained at - 80 °C until analyses were performed.

2.3. DNA extraction and shotgun metagenomic sequencing

Nucleic acid extraction was performed at the UC San Diego Microbiome Core utilizing previously published protocols (<https://doi.org/10.1186/s40168-021-01083-0>). Metagenomic libraries were prepared with KAPA HyperPlus kits (Roche Diagnostics, USA) following manufacturer's instructions. Sequencing was performed on an Illumina NovaSeq 6000 sequencing platform with NovaSeq 6000 SP Reagent Kit v1.5 and paired-end 150 bp cycle chemistry.

Sequencing and Statistical Analysis.

Host-derived reads were removed using kneadData (v0.11.0) and taxonomic classifications were obtained using kraken2 (v2.1.1). Additional analyses were performed in the R statistical environment (v3.6.3) using the 'phyloseq' (v1.30.0) package. Gut microbiota variability longitudinally was quantified using Jensen-Shannon divergence and comparisons were performed using a Wilcoxon rank-sum test. Spearman correlation was used to assess the relationship between gut microbiota diversity using the Chao1 index and microbiota variation.

3. Results

3.1. Participant characteristics

Participant characteristics are described in Table 1. Participants

Table 1
Participant characteristics.

Subject	Age at Time of HCT	Indication for Allogeneic HCT	HCT Source	Conditioning Regimen	GvHD Prophylaxis	Maximum Gut GvHD Stage	Maximum GvHD Stage (S, L)	Alive or Deceased
1	3 years	HR-ALL	6/6 MRD PBSC	Cyclophosphamide, Cytarabine, TBI	Tacrolimus, Steroids	1	0, 0	Alive
2	15 months	Wiskott-Aldrich Syndrome	10/10 MUD BM	Busulfan, Fludarabine, ATG	Tacrolimus, Steroids	2	1, 0	Alive
3	19 years	HR-ALL, CNS 3	4/6 UCB	Cyclophosphamide, Cytarabine, TBI	Tacrolimus, Steroids	4	1, 0	Deceased at 17-months post-HCT
4	13 years	HR-ALL Ph+ status-post-CD-19 directed CAR-T therapy	10/10 MSD PBSC	Cyclophosphamide, Cytarabine, TBI	Tacrolimus, Steroids	0	0, 0	Alive
5	5 years	HR-ALL	10/10 MUD PBSC	Cyclophosphamide, Cytarabine, TBI	Tacrolimus, Methotrexate, Steroids	4	2, 0	Deceased at 22-months post-HCT
6	10 years	HR-ALL	5/6 UCB	Cyclophosphamide, Cytarabine, TBI	Tacrolimus, Steroids	0	1, 0	Alive
7	11 years	HR-ALL	10/10 MUD	Cyclophosphamide, Cytarabine, TBI	Tacrolimus, Steroids	2	3, 0	Alive
8	15 years	X-Linked Lymphoproliferative Disease	10/10 MUD BM	Cyclophosphamide, Busulfan	Tacrolimus, Methotrexate, Steroids	2	1, 0	Alive
9	4 months	DOCK 8 Deficiency	10/10 MUD BM	Cyclophosphamide, Busulfan	Tacrolimus, Methotrexate, Steroids	2	4, 2	Alive

Abbreviations: HCT, hematopoietic cell transplantation; GvHD, graft-versus-host disease; (S, L), (skin, liver); HR-ALL, high-risk acute lymphoblastic leukemia; Ph+, Philadelphia-like; CNS, central nervous system; CAR-T, chimeric antigen receptor T cell; MRD, matched related donor; MSD, matched sibling donor, PBSC, peripheral blood stem cells; BM, bone marrow; UCB, umbilical cord blood; TBI, total body irradiation.

ranged in age from 4 months to 19 years old at the time of allogeneic HCT. The indication for allogeneic HCT included three non-malignant conditions: Wiskott-Aldrich syndrome, X-linked lymphoproliferative disease, DOCK8 deficiency (participants 2, 8, and 9 respectively). Six children were transplanted for high-risk acute lymphoblastic leukemia (ALL) one of whom received CD19-directed chimeric antigen receptor T cell (CAR-T) therapy with Tisagenlecleucel as consolidation therapy prior to HCT (Participant 4).

Two participants (Participant 3 and 5) experienced severe stages of gut GVHD, namely stages 3 or 4, at least once within the first year post-HCT and were categorized as 'gut GVHD' participants. The remaining seven participants were categorized as 'controls.' Two participants did not experience gut GVHD (Participants 4 and 6), five participants experienced mild stages 1 or 2 of gut GVHD and 2 control participants, (Participants 7 and 9), experienced severe skin GVHD with a maximum stage of 3 and 4, respectively without stages 3 or 4 gut GVHD. Moreover, two participants (Participants 3 and 5) experienced stage 4 gut GVHD during the study period and unfortunately both died after the study period from transplantation-related complications.

3.2. Longitudinal changes in the gut microbiota with corresponding GVHD staging following pediatric allogeneic hematopoietic cell transplantation

Nine participants provided serial stool collections for a total of 117 time points (Fig. 1A). Due to unforeseen interruptions from the COVID-19 pandemic, some specimens were not collected at the initially proposed time points, however, every attempt was made to maintain adherence to the proposed collection schedule. The days of actual collection are noted in the figures.

Fig. 1. A shows the relative abundance of the bacterial genera in stool samples collected longitudinally at each time point for every participant. All participants had stool specimens collected to at least Day + 90 post-HCT. Three participants, namely Participants 3, 7 and 8, provided stools specimens to at least Day + 300 post-HCT. The Chao1 index (or alpha diversity) is displayed below each time point on a gray-scale gradient with black representing a stool sample with high diversity and light gray

representing a stool sample with low diversity. Participants 6, 7 and 8 maintained a high Chao1 index throughout the study period despite various antibiotic exposures. Most participants entered their conditioning regimen with relatively diverse bacterial gut microbiotas as depicted by a high Chao1 index, except for Participants 2 and 3.

When comparing the relative abundance changes from the pre-HCT sample compared to the peri-engraftment period up to Day + 21, Participants 6 and 7 were able to maintain a similar bacterial relative abundance distribution to their pre-HCT sample. Participant 2 displays a co-domination with *Bifidobacterium* and *Enterocloster* on Day 0 (day of HCT) likely due to the participants conditioning regimen but was able to recover their pre-HCT microbiota signature by Day + 7 followed by increasing alpha diversity (e.g., increasing Chao 1 index) over time. Participant 9 had a moderate alpha diversity prior to transplantation, however, this participant's post-transplantation stool showed time-points with near mono-domination with *Enterococcus* or low diversity with *Enterobacter*, *Escherichia* and/or *Bacteroides* suggesting inability to recover one's pre-HCT microbiota diversity. The other 5 participants showed pronounced deviations in the bacterial relative abundance in the peri-engraftment period and were unable to recover their baseline gut microbiota signature during the study period.

Antibiotic exposures during the study are noted in red acronyms below the Chao 1 index (see antibiotic figure legend in Fig. 1A). Participant 1 was diagnosed with an *E. coli* urinary tract infection on Day 0 treated with cefepime. Participant 2 was given cefepime for y diarrhea then transitioned to oral vancomycin for *Clostridium difficile* enteritis. Participant 2 experienced a *Pseudomonas* central line infection on Day + 195 which was treated with cefepime. Participant 3's infectious history is described in greater detail in Fig. 1C. Participant 4 experienced an infected preauricular cyst with *Enterococcus* which was initially treated with piperacillin/tazobactam followed by amoxicillin/clavulanic acid. Participant 5 developed methicillin-sensitive *Staphylococcus aureus* bacteremia on Day + 7 and completed a 21-day course of oxacillin. Participants 6 and 7 did not experience any bacterial infections during the study period. Participant 8 developed a co-infection with methicillin-sensitive *Staphylococcus aureus* and *Streptococcus viridans*

Fig. 1. Panel A shows the relative abundance of bacterial genera in stool samples collected longitudinally following allogeneic hematopoietic cell transplantation (HCT). The x-axis displays the pre-HCT stool obtain prior to or on the first day of the conditioning regimen represented as Day - 1 followed by post-HCT stool collections numbered by day post-HCT. The y-axis displays the relative abundance of various bacterial genera at each time point. Bacterial diversity (Chao1) is shown along the bottom. Administration of antibiotics is indicated in red text along the bottom (see antibiotic key; IV, intravenous; PO, by mouth). * indicates gut GVHD participants. Panel B shows the longitudinal time course of graft-versus-host disease (GVHD) by organ system and stage for each participant. Each circle represents a sample and data collection time point per study protocol. The width of each participant's bar represents the duration of their involvement in the study. Each dot represents the GVHD organ system and stage at the time of specimen collection. Organ systems are colored coded with red representing skin GVHD, blue representing gut GVHD and green representing liver GVHD. Open circles are used to represent mild GVHD defined as stages 1–1 and severe stages as 3–4 are displayed as closed circles. Participants were assessed for GVHD weekly while admitted to the hospital for any indications and during specimen collection times per study protocol. Arrow represents the post-HCT day the intestinal biopsy was performed. Not all participants met clinical criteria for a biopsy throughout the study period. Panel C demonstrates an example of a participant (Participant 3) who developed Stages 3–4 gut GVHD following allogeneic HCT for high-risk acute lymphoblastic leukemia. The time course of the relative bacterial abundance of stool specimens collected longitudinally along with corresponding antibiotic therapies, GVHD organ system and stage and infections up to Day 395 post-transplantation are shown.

bacteremia which was treated with initially with cefepime then switched to oxacillin. Participant 9 developed recurrent central line infections with *Serratia marcescens* (Days +175 and +270). All other timepoints which have an antibiotic exposure noted were given for neutropenic fever, hemodynamic instability or at the discretion of the treating physician based on the participant's clinical course without a documented pathogen.

The longitudinal time course of GVHD by stage is described for each participant at the defined study time points in Fig. 1B. All participants except for Participant 1 and 4 experienced GVHD throughout their study period. Participants 3 and 5 experienced severe gut GVHD at least 3 times throughout their post-transplantation course. Participants 7 and 9 experienced severe skin GVHD at least once following transplantation. Additionally, Participant 9 was the only participant to experience mild liver GVHD during the study period.

Fig. 1. C highlights the post-transplantation course of Participant 3 whom experienced recurrent episodes of severe gut GVHD. As noted above, Participant 3 displayed a pre-HCT stool microbiota with a low bacterial diversity followed by near mono-dominance of *Escherichia* after conditioning. Participant 3 received cefepime during the peri-engraftment period due to neutropenic fever (Days +7 and +14) and again on Day + 49 for neutropenic fever with a corresponding urinary tract infection with *Escherichia coli* and *Staphylococcus epidermidis* bacteremia; later narrowed to monotherapy with intravenous vancomycin. Moreover, Participant 3 also developed *Salmonella* enteritis at Day + 395. In parallel, this participant also developed recurrence viremia and viuria due to Adenovirus while on cidofovir and cytomegalovirus reactivation between Days + 49 to + 70 which responded to ganciclovir treatment. Ultimately, this participant was diagnosed with biopsy-proven transplantation-associated thrombotic microangiopathy (TA-TMA) and started placed eculizumab therapy (monoclonal antibody to inhibit the complement pathway). Unfortunately, Participant 3 subsequently developed severe sepsis due to multi-drug resistant *Escherichia coli* bacteremia which progressed to respiratory failure requiring extracorporeal membrane oxygenation (ECMO) leading to death.

3.3. Longitudinal bacterial gut microbiota variance in “gut GVHD” (Stages 3–4) versus “controls” (Stages 0–2)

We sought to compare the longitudinal variance of the bacterial gut microbiota over time between “controls” versus “gut GVHD” participants using the Jensen-Shannon Divergence (JSD). The JSD was used to determine whether there is greater bacterial microbiota variation within an individual than between individuals. We found that GVHD participants experienced greater variation in their bacterial gut microbiota diversity over time than the control participants (Fig. 2, $p = 0.007$), however, this was a descriptive study and thus a large cohort would be needed to determine if this relationship persists.

Relationship between Pre-HCT Bacterial Gut Microbiota Diversity to Post-HCT Microbiota Variance.

Given that “gut GVHD” participants showed a greater variation in the gut microbiota diversity over time, we sought to determine if there was a

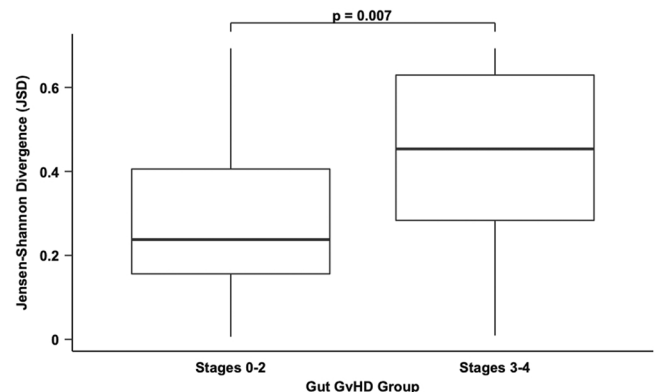


Fig. 2. Boxplot shows the longitudinal variance of the gut microbiota between participants with mild (stages 0–2) versus severe (stages 3–4) gut GVHD. Longitudinal variance was calculated using Jensen-Shannon Divergence (JSD), with higher values representing greater variability in microbiota composition over time. *Wilcoxon rank-sum test was used for the comparison (** $p < 0.01$).

relationship between pre-HCT gut microbiota diversity and post-transplantation gut microbiota variation. Pre-HCT Chao1 index (alpha diversity) was compared to their post-HCT variance over time using the JSD for each participant. In Fig. 3, we observed a trend towards an inverse relationship, namely low pre-HCT alpha diversity may correlate with high post-HCT microbiota variation, however, this did not reach statistical significance ($p = 0.05$). Importantly, the outliers in this graph, namely Participants 3 and 9, align with the inverse relationship being explored in this comparison, however, the remaining participants do not support this hypothesis.

4. Discussion

In this small prospective observational study, we longitudinally characterize the gut microbiota of pediatric allogeneic HCT participants. We demonstrate that children who develop stage 3 or 4 gut GVHD during their HCT course have greater variation in their gut microbiota over time compared to children who do not develop gut GVHD or have milder forms (Stage 1–2) gut GVHD (Fig. 2, $p = 0.007$). Furthermore, we found that pre-HCT gut microbiota diversity trended towards an inverse relationship with gut microbiota stability over time, however, this did not reach statistical significance (Fig. 3, $p = 0.05$). These findings begin to provide insight into the bacterial microbiota changes that occur following allogeneic HCT in pediatric patients. Given this is a descriptive study and a small participant cohort, these findings will require validation in large pediatric cohort studies in the future.

Most participants had a diverse gut microbiome prior to or on the first day of their conditioning regimen (7 out of 9 participants). One of two participants with notably less microbiota diversity had undergone multiple rounds of chemotherapy and antibiotics prior to HCT and the second participant was a 15-month-old with Wiskott-Aldrich syndrome.

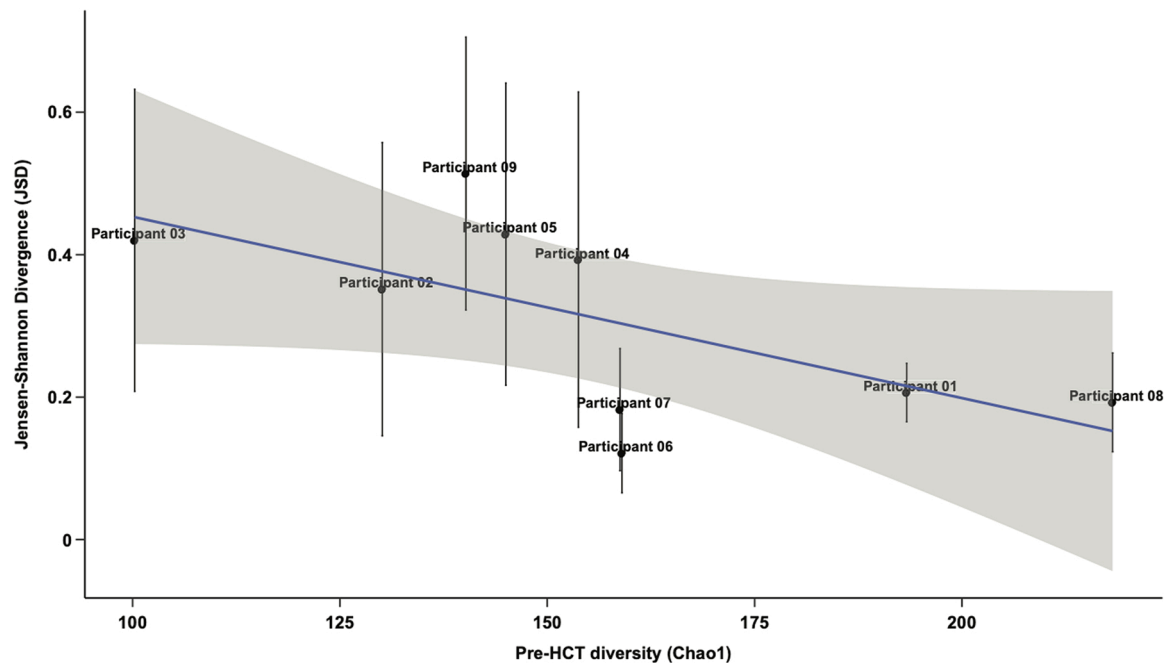


Fig. 3. Graph above shows the relationship between the pre-transplantation bacterial diversity of each participant and the participant's microbiota variability over time post-transplantation. The x-axis shows the pre-HCT Chao1 alpha diversity measure for the stool specimen which was obtained prior to or on the first day of the conditioning regimen. The y-axis represents the mean Jensen-Shannon Divergence (JSD) of the gut microbiota for the same individual post-transplantation to demonstrate the change in bacterial diversity over time. Vertical black lines denote standard deviations, blue line and shaded area show the fitted linear regression and 95 % confidence intervals. Spearman correlation ρ and associated p-value are shown in parentheses.

Supplemental Fig. 1: This figure shows that there is no difference in the number of read counts between pre-HCT and Day 0 sampling time points confirming the relative bacterial abundance is not due to a sampling error.

Consistent with a study of adult HCT patients, most of our participants (7 out of 9) experienced significant deviations in bacterial relative abundance in the peri-engraftment period and were unable to recover their baseline microbiota up to Day + 21 (Kusakabe et al., 2020). Following neutrophil engraftment, three participants, namely Participants 6–8, were able to maintain a highly diverse microbiota signature with a *Bacteroides* dominance despite antibiotic exposures, conditioning and development of mild GVHD. Studies have shown that *Bacteroides* predominance is associated with grade II–IV gut GVHD (Biagi et al., 2019). This contrasts with our study where Participants 6 and 7 either maintained or at times had high relative abundances of *Bacteroides* following transplantation but did not develop any stage of gut GVHD. This may suggest that predominance of a particular bacterial species may not correlate with GVHD risk in all patients. Further investigations into what clinical and physiological factors contribute to the stability of the gut microbiota after allogeneic HCT in children requires further exploration.

Participant 3's clinical course and gut microbiota analysis provides insight into the longitudinal gut microbiota perturbations that occur in a patient with severe and recurrent gut GVHD. Interestingly, Participant 3's pre-transplantation stool specimen displayed the lowest alpha diversity of our cohort (Fig. 1a and Fig. 3) which is consistent with similar children with exposure chemotherapy, radiation, and antibiotic before HCT (Elgarten et al., 2022). Following conditioning, this participant displayed a pronounced mono-domination of *Escherichia* during the peri-engraftment period which has been associated with gut GVHD in both mouse models and human studies suggesting (Eriguchi et al., 2012; Heimesaat et al., 2010; Peled et al., 2020; Elgarten et al., 2022). Participant 3 experienced a brief recovery period with increasing alpha diversity; however, this participant subsequently had a recurrence of Stage 3–4 gut GVHD characterized by a swift domination with *Enterococcus* and *Lactocaseibacillus* genera. Interestingly, domination with *Enterococcus* has been associated with increased risk of gut GVHD (Peled et al., 2020; Stein-Thoeringer et al., 2019; Simms-Waldrup et al., 2017)

and bacteremia (Holler et al., 2014) which is consistent with Participant 3's post-HCT complications. Lastly, Participant 3 also experienced recurrence episodes of severe gut GVHD along with increased gut microbiota variation and recurrent infections (Fig. 3). One possible explanation is that acute changes in the bacterial genera may lead to a microenvironment within the gut that hinders or does not promote mucosal healing allowing resident organisms to cause recurrent infections as seen in Participant 3.

Antibiotic exposures during the peri-engraftment period may affect a patient's ability to maintain or recover their pre-HCT microbiota diversity. Our cohort represents a population that did not received prophylactic antibiotics unless clinically indicated (i.e., fever, hemodynamic instability). Most of our participants (7 out of 9) experienced pronounced deviations in the bacterial relative abundance in the peri-engraftment period and were unable to recover their baseline gut microbiota signature during the study period. This is consistent with other studies, Biagi et al. compared pre-HCT stool specimens to post-HCT from ten pediatric allogeneic HCT patients, five of whom experienced acute GVHD (four patients with skin GVHD and one patient with gut and skin GVHD) (Biagi et al., 2015). These investigators showed that HCT can lead to a temporal 30 % reduction in the alpha diversity followed by a return to their baseline by Day + 65 to + 103 post-transplantation. However, 2 out of 9 of our participants were able to maintain their alpha diversity with stable bacterial relative abundances throughout transplantation. Biagi et al. utilized levofloxacin prophylaxis for gut decontamination prior to the start of the conditioning regimen and continued until neutrophil engraftment. The rapid return of the alpha diversity post-transplantation was likely impacted by the discontinuation of broad-spectrum antibiotics. In contrast, our study did not utilize a prophylactic antibiotic strategy during periods of neutropenia, and this may contribute to why some participants were able to maintain their alpha diversity following their conditioning regimen and after HCT (Fig. 1A, Chao1 Index). Since decreased microbiota diversity

after HCT is associated with GVHD and transplantation-related mortality (Peled et al., 2020), judicious antibiotic use in the pre- and peri-engraftment period is critical.

Unfortunately, both participants who developed severe gut GVHD during the study died after the study period (>1-year post-HCT) from infectious complications while on immunosuppressive therapy for chronic GVHD (Participant 3 for gut GVHD and Participant 5 for liver GVHD). Patients who experience steroid-refractory/-resistant gut GVHD have a mortality of 80–100 % within 1 year after transplantation (Castilla-Llorente et al., 2014). Our findings of increased variation in the gut microbial diversity in these two participants are consistent with a prior report that pediatric patients with gut GVHD have greater inter-individual gut microbiota variability than those without GVHD (Biagi et al., 2019). We postulate that children with severe gut GVHD may have less resilient gut microbiotas that contribute to their inability to either maintain or return to their baseline diversity. We were unable to show a significant correlation between pre-HCT diversity and post-HCT variation in pediatric participants (Fig. 3) likely due to our small sample size.

Strengths of our study include a prospective, descriptive design with longitudinal stool sampling of pediatric allogeneic HCT participants, allowing for a detailed description of the gut microbiota changes that occur within an individual child. Limitations to the study include a small sample size, wide pediatric age range, and few participants with severe gut GVHD. Additionally, we analyzed children with malignant and non-malignant indications for HCT together. In pediatric acute lymphoblastic leukemia patients, gut dysbiosis during the pre-transplantation periods has been previously described which has been attributed to multiple rounds of chemotherapy and antibiotic courses (Oldenburg et al., 2021; Thomas et al., 2020). In contrast, many children with non-malignant diseases have not received any chemotherapy prior to HCT and limited antibiotic exposure. Furthermore, we included pediatric participants with a broad age range which included children 4-months-old to 19-years-old. Studies have shown that the bacterial gut microbiota does not stabilize until approximately 3 years old thus some variation seen following allogeneic transplantation may be attributed to age-related differences (Milani et al., 2017; Derrien et al., 2019). Larger multi-center cohort studies are necessary to address these limitations.

5. Conclusion

In conclusion, we characterized the longitudinal changes in the bacterial gut microbiotas of nine pediatric HCT participants and demonstrated that those children with stages 3 or 4 gut GVHD have significantly greater variation in their gut microbiota diversity over time. Our data provides a in-depth descriptive characterization of pediatric patients after allogeneic HCT and can be used to develop larger prospective studies to begin elucidate what factors contribute to a child's ability to maintain a stable, diverse gut microbial community following allogeneic HCT. Furthermore, our data began to explore whether the pre-transplantation gut microbial diversity was correlated with post-HCT gut bacterial variability over time. This could be an important factor to investigate in future prospective studies as a possible marker of post-transplant GVHD-risk and to help decide guide which patients would benefit most from gut microbiota-directed therapeutic approaches in the future.

Data availability

Data will be made available on request.

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and sample preservation. A.N. Gray acknowledges support by the National Institute of Health (NIH) T32 Developmental Hematology Training grant (T32 HL086345). The UC San Diego Microbiome Core performed sample extraction and Illumina library preparation. This publication includes data generated at the UC San Diego IGM Genomics Center utilizing an Illumina NovaSeq 6000 that was purchased with funding from a National Institutes of Health SIG grant (#S10 OD026929).

Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at doi:10.1016/j.ijmm.2023.151580.

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